



Technical Service Bulletin

Technical Service Bulletin: TSB210091

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Aftertreatment DPF Regen Analyzer (ARA) Available to Diagnose: Black Smoke, Regeneration Excessive, Stationary Regeneration Will Not Complete, Turbocharger Noise

Aftertreatment DPF Regen Analyzer (ARA) Available to Diagnose: Black Smoke, Regeneration Excessive, Stationary Regeneration Will Not Complete, Turbocharger Noise

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- B6.7 CM2350 B121B
- B6.7 CM2450 B155B
- ISB CM2150
- ISB6.7 CM2250
- ISB6.7 CM2350 B101
- ISC CM2150
- ISC8.3 CM2250
- ISL CM2150
- ISL9 CM2250
- ISL9 CM2350 L101
- ISX CM871
- ISX12 CM2350 X102
- ISX12/ISX11.9 CM2250
- ISX15 CM2250
- ISX15 CM2350 X101
- L9 CM2350 L116B
- L9 CM2350 L123B
- L9 CM2450 L126B
- X12 CM2350 X119B
- X12 CM2450 X137B
- X15 CM2350 X114B
- X15 CM2350 X116B

- X15 CM2450 X124B
- X15 CM2450 X134B

Description of Change

The Aftertreatment Regeneration Analyzer (ARA) test is available in the INSITE™ electronic service tool. This diagnostic test was developed to improve troubleshooting of engine air management and aftertreatment systems for most EPA 2007+ Cummins® Heavy Duty and Midrange engines in on-highway applications.

Issue

Engine experiences the following Fault Codes or Symptoms:

- 1921, 1922, 1981, 2387, 2754, 3361, 3375, 3376, 4612
- Black Smoke, Dynamic Engine System Analysis Will Not Complete, Regeneration Excessive, Stationary Regeneration Will Not Complete, Turbocharger Noise

Verification

- The ARA test is used when troubleshooting certain fault codes or engine symptoms. The ARA performs a partial Stationary Regeneration to bring the engine and aftertreatment up to temperature and then datalogs various parameters. This information is analyzed to determine what components or systems may be contributing to the symptom being investigated. The test gives a PASS/FAIL/INSPECT judgment and recommends additional troubleshooting for systems and/or components that do **not** have a PASS result.
- The ARA test should be performed when directed by troubleshooting to verify the root cause of failure is identified and repaired the first time without unnecessary component replacements. The ARA test is currently indicated to be used by Guidanz®, EDS, and QuickServe® Online troubleshooting for multiple ECM Fault Codes and engine symptoms.

Resolution

- Perform ARA when directed by published troubleshooting.
- Reference the appropriate Fault Code Troubleshooting procedure. See corresponding Service Manual. Reference Procedure 014-027 in Section 14.

Service Instructions

- Cummins Inc. has taken several actions to encourage use of the ARA. Training is offered on the test as part of Cummins® Virtual College (CVC) course 1443, Turbocharger Shop Talk.

Note : https://cst.cummins.com/ServiceTraining/tech_hp.php
(https://cst.cummins.com/ServiceTraining/tech_hp.php)

- Cummins Inc. has posted a short video about the ARA in the Cummins® YouTube channel.

Note : <https://www.youtube.com/watch?v=LxIGtPaCTPM>
(<https://www.youtube.com/watch?v=LxIGtPaCTPM>)

- Cummins Inc. has posted a short video about how the ARA replaces the sector gear hand check for diagnosing a turbocharger.

Note : http://tsb.cumminsvirtualcollege.com/TSB210091_en.aspx
(http://tsb.cumminsvirtualcollege.com/TSB210091_en.aspx)

- Articles regarding the ARA are available on the ServiceLine website.

Note : <https://serviceline.cummins.com/> (<https://serviceline.cummins.com/>)

- An SRT has been released to reimburse for performing the test. The SRT can be claimed when the ARA is directed and performed during a warrantable repair. EDS/Guidanz® will give the SRT if the test is directed by troubleshooting; the SRT can also be searched on QSOL or in your claim filing system. INSITE™ electronic service tool sends the data collected and results of the ARA back to Cummins®. This data is being matched to warranty claims on failed Diesel Oxidation Catalysts (DOC), Diesel Particulate Filters (DPF), turbochargers, turbocharger actuators, and other components. Cummins® reserves the right to make claim adjustments where the ARA passes a claimed component or where the ARA is directed by troubleshooting and **not** performed.

Document History

Date	Details
2021-4-20	Module Created
2021-9-13	Adding link to new video explaining how the ARA replaces the sector gear hand check for diagnosing a turbocharger.

Last Modified: 14-Sep-2021
